

Localization and mobility edges in one-dimensional deterministic potentials

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In this paper, we study the localization properties of the wave function of a one-dimensional tight-binding electron moving in an asymptotic periodic potential, $V_n = \lambda \cos(2\pi Qn + \pi\alpha n^\nu)$, where n is the site index and $0 < \nu < 1$. For Q rational, the electronic energy band consists of many subbands, and the number of subbands is determined by Q . For $\lambda < 2$, there are two mobility edges where the eigenstates at the subband center are all extended, whereas the subband-edge states are all localized in every subband. We develop some heuristic arguments to calculate exactly the mobility edges for this model and carry out numerical work to study the localization properties of the model. Our theoretical results are essentially in exact agreement with the numerical results. We calculate the critical exponents δ and β at mobility edges. We also study the nature of the localized, extended eigenstates and mobility edges of this system as a function of λ , α , and ν .

I. INTRODUCTION

Since Anderson pointed out that the states of an electron moving in a random potential would be localized,¹ the features of thin states have been extensively investigated in recent literature for a wide class of potentials.²⁻¹⁴ Several investigations have been carried out for a one-dimensional nearest-neighbor tight-binding model

$$U_{n+1} + U_{n-1} + V_n U_n = E U_n, \quad (1)$$

where the strength of the nearest-neighbor hopping has been set equal to unity with no loss of generality. In Eq. (1), U_n is the amplitude of the electronic wave function at the n th lattice site, V_n is the on-site diagonal potential, and E is the electronic eigenenergy. The choice of the diagonal potential V_n completely defines the model and the physical problem being studied. For example, a periodic V_n is the usual Bloch case whereas Eq. (1) with a random V_n is the Anderson model. It is well known that a V_n periodic in n has all its states extended, whereas a wide class of random V_n produces^{2,3} all localized states in one dimension. Intermediate between the periodic and the random situations is the incommensurate case, with a periodic V_n having a period incommensurate with the underlying lattice. The best studied incommensurate potential is Harper's model,³⁻¹¹ where $V_n = \lambda \cos(2\pi Qn)$ with Q being an irrational number. This model undergoes a metal-insulator transition such that the states for $\lambda < 2$ are extended, those for $\lambda > 2$ are localized, and the state at $\lambda = 2$ becomes critical (neither localized nor extended).

In recent years, a class of general model potentials has been introduced and extensively studied which exhibits an energy-dependent transition (mobility edge) in one-dimensional systems. These potentials are deterministic and yet different in their behavior from random potentials in the sense that, in one dimension, all eigenfunctions are not localized. Instead, there appear metal-insulator transitions separating extended and localized states. The potential¹²⁻¹⁵ $V_n = \lambda \cos(\pi\alpha n^\nu)$ has been of

much interest in recent years. Das Sarma, He, and Xia¹⁵ have calculated the wave function, electronic density of states, and Lyapunov exponent for the above model in the regime $0 < \nu < 1$, which shows an energy-dependent transition at $E = \pm|2 - \lambda|$.

In this paper, we study a class of more general one-dimensional deterministic and aperiodic models described by the equation

$$V_n = \lambda \cos(2\pi Qn + \pi\alpha n^\nu), \quad (2)$$

where α and Q are positive real numbers and λ is the strength of the potential. The positive exponent ν in Eq. (2) is taken to lie between 0 and 1. When $\alpha = 0$ and Q is rational, we get back the periodic Bloch model, whereas it has been shown¹⁴ that for $Q = 0$, α irrational, and $\nu \geq 2$ the pseudorandom model defined by Eqs. (1) and (2) becomes equivalent to a corresponding random Anderson model. For $\alpha = 0$ and Q irrational, Eq. (2) defines Harper's model, whereas for $Q = 0$ and α irrational Eq. (2) is the model studied by Das Sarma, He, and Xia.¹⁵ In this paper, we study the localization properties of the model (2) for $Q \neq 0$, especially for $Q = \frac{1}{2}$ and $\frac{1}{3}$.

The rest of the paper is organized in the following manner. In Sec. II, we present some heuristic arguments. In Sec. III, we present our numerical results, comparing them with theory. We provide the conclusion in Sec. IV.

II. HEURISTIC ARGUMENTS

Since the cases with λ and $-\lambda$ are equivalent, we assume $\lambda > 0$ hereafter. From Eq. (2) we can write

$$V_{n+1/Q} - V_n = -2\lambda \sin \left\{ \frac{\pi\alpha}{2} \left[\left(n + \frac{1}{Q} \right)^\nu - n^\nu \right] \right\} \\ \times \sin \left\{ 2\pi Qn + \frac{\pi\alpha}{2} \left[n^\nu + \left(n + \frac{1}{Q} \right)^\nu \right] \right\} \quad (3)$$

and

$$\left[n + \frac{1}{Q} \right]^v - n^v \simeq \frac{v}{Qn^{1-v}} \rightarrow 0 \text{ as } n \rightarrow \infty \text{ and } 0 < v < 1, \quad (4)$$

implying

$$V_{n+1/Q} - V_n \rightarrow 0 \text{ as } n \rightarrow \infty. \quad (5)$$

Thus in the limit $n \rightarrow \infty$, the potential (2) tends to a periodic function with period $1/Q$. In the remainder of this section, the method of argument is similar to that used by Das Sarma, He, and Xia¹⁵ in studying the localization properties of a class of one-dimensional slowly varying potentials. We discuss the two cases of $Q = \frac{1}{2}$ and $\frac{1}{3}$.

A. Case I: $Q = \frac{1}{2}$

For $Q = \frac{1}{2}$, the potential (2) is an asymptotic periodic potential with period 2. Then, as $m \rightarrow \infty$, we can write

$$V_{2m} = \lambda \cos(\beta) \text{ and } V_{2m+1} = \lambda \cos(\beta + \pi) \quad (6)$$

with β an arbitrary real number.

Equation (1) can be rewritten as

$$U_{2m-2} + U_{2m} - C_2 U_{2m-1} = 0, \quad (7)$$

$$U_{2m-1} + U_{2m+1} - C_1 U_{2m} = 0, \quad (8)$$

$$U_{2m} + U_{2m+2} - C_2 U_{2m+1} = 0, \quad (9)$$

with

$$C_1 = E - V_{2m} \text{ and } C_2 = E - V_{2m+1}. \quad (10)$$

In Eqs. (7)–(9), we eliminate the U_{2m+1} and U_{2m-1} , and get

$$U_{2m+2} + U_{2m-2} - C U_{2m} = 0, \quad (11)$$

where

$$C = C_1 C_2 - 2. \quad (12)$$

We write

$$U_{2m} \sim z^m, \quad (13)$$

where z is a complex quantity.

Equation (11) becomes

$$z^2 - C z + 1 = 0 \quad (14)$$

with the complex solutions

$$z_{1,2} = \frac{1}{2}(C \pm \sqrt{C^2 - 4}). \quad (15)$$

From Eq. (15) we conclude that the amplitude is complex and $|z_1| = |z_2| = 1$ or extended if $|C| < 2$, whereas it is real or localized if $|C| > 2$.

From the expressions (6), (10), and (12), we get

$$C = E^2 - \lambda^2 \cos^2(\beta) - 2. \quad (16)$$

When E and λ are fixed, we get

$$C_{\max} = E^2 - 2 \text{ and } C_{\min} = E^2 - \lambda^2 - 2. \quad (17)$$

The conditions for extended and localized solutions are

$$C_{\max} = \begin{cases} E^2 - 2 < 2, & \text{i.e., } |E| < 2 \text{ (extended)}, \\ E^2 - 2 > 2, & \text{i.e., } |E| > 2 \text{ (localized)}, \end{cases} \quad (18)$$

$$(19)$$

and

$$C_{\min} = \begin{cases} E^2 - \lambda^2 - 2 > -2, & \text{i.e., } |E| > \lambda \text{ (extended)}, \\ E^2 - \lambda^2 - 2 < -2, & \text{i.e., } |E| < \lambda \text{ (localized)}. \end{cases} \quad (20)$$

$$(21)$$

Thus, for $\lambda < 2$, there are four metal-insulator transitions in the system of $Q = \frac{1}{2}$ at the mobility edges $E_{c_1} = \pm\lambda$ and $E_{c_2} = \pm 2$, and for $\lambda > 2$, there is no extended state.

B. Case II: $Q = \frac{1}{3}$

Similarly, the potential (2) is an asymptotic periodic potential with period 3 for $Q = \frac{1}{3}$. Then as $m \rightarrow \infty$, we can write

$$V_{3m} = \lambda \cos(\beta), \quad V_{3m+1} = \lambda \cos\left[\beta + \frac{2\pi}{3}\right],$$

and

$$(22)$$

$$V_{3m+2} = \lambda \cos\left[\beta + \frac{4\pi}{3}\right]$$

with β an arbitrary real number.

From Eq. (1), we get the equation

$$U_{3m+3} + U_{3m-3} - C U_{3m} = 0, \quad (23)$$

where

$$C = C_1 C_2 C_3 - C_1 - C_2 - C_3 \quad (24)$$

and

$$C_1 = E - V_{3m}, \quad C_2 = E - V_{3m+1}, \text{ and } C_3 = E - V_{3m+2}. \quad (25)$$

We write $U_{3m} \sim z^m$, where z is a complex quantity. Thus we get an equation and solutions the same as Eqs. (14) and (15), respectively. Then the condition for extended states is $|C| < 2$. From the expressions (22), (24), and (25), we get

$$C = E^3 - \frac{3}{4}\lambda^2 E - 3E - \lambda^3 \cos(\beta) [\cos^2(\beta) - \frac{3}{4}]. \quad (26)$$

When E and λ are fixed, we get

$$C_{\max} = E^3 - \frac{3}{4}\lambda^2 E - 3E + \frac{\lambda^3}{4}, \quad (27)$$

$$C_{\min} = E^3 - \frac{3}{4}\lambda^2 E - 3E - \frac{\lambda^3}{4}. \quad (28)$$

Then the condition $|C| < 2$ becomes

$$f_1(E) = C_{\max} - 2 = E^3 - \frac{3}{4}\lambda^2 E - 3E + \frac{\lambda^3}{4} - 2 < 0, \quad (29)$$

$$f_2(E) = C_{\min} + 2 = E^3 - \frac{3}{4}\lambda^2 E - 3E - \frac{\lambda^3}{4} + 2 > 0. \quad (30)$$

Because the criteria of the equations $f_1(E)=0$ and $f_2(E)=0$ are the same and equal to

$$\Delta = \left[\frac{\lambda^3}{8} - 1 \right]^2 + \left[-\frac{\lambda^2}{4} - 1 \right]^3$$

$$= -\frac{\lambda^2}{16} (3\lambda^2 + 4\lambda + 12) < 0 \text{ for every } \lambda, \quad (31)$$

the roots of equations $f_1(E)=0$ and $f_2(E)=0$ are real. Let x_1, x_2, x_3 and y_1, y_2, y_3 be the three roots of $f_1(E)=0$ and $f_2(E)=0$, respectively. Then

$$x_1 = -y_1 = 2\sqrt{(1 + \frac{1}{4}\lambda^2)} \cos(\theta), \quad (32)$$

$$x_2 = -y_3 = -2\sqrt{(1 + \frac{1}{4}\lambda^2)} \cos\left[\theta - \frac{\pi}{3}\right], \quad (33)$$

$$x_3 = -y_2 = -2\sqrt{(1 + \frac{1}{4}\lambda^2)} \cos\left[\theta + \frac{\pi}{3}\right], \quad (34)$$

with

$$\cos(\theta) = \frac{(1 - \frac{1}{8}\lambda^3)}{\sqrt{(1 + \frac{1}{4}\lambda^2)^3}}. \quad (35)$$

Then for $\lambda < 2$ the conditions for the extended state become

$$|E| < E_1, \quad E_2 < E < E_3, \quad \text{and} \quad -E_2 > E > -E_3, \quad (36)$$

with $E_1 = |x_3| = |y_2|$, $E_2 = |x_2| = |y_3|$, $E_3 = |x_1| = |y_1|$, and $E_1 < E_2 < E_3$. For $\lambda > 2$ there is no value of E that satisfies the conditions for the extended state and all states are localized.

So that, for $Q = \frac{1}{3}$ and $\lambda < 2$, there are six mobility edges located at $\pm E_1$, $\pm E_2$, and $\pm E_3$. In Table I we list the mobility edges for different λ from $\lambda=0.2$ to 2.0 .

III. NUMERICAL RESULTS

Within the nearest-neighbor approximation, the tight-binding Hamiltonian is a tridiagonal matrix. We use

Dean's method¹⁶ and the direct diagonalization method to calculate the eigenenergies and eigenfunctions of the model Hamiltonian defined by Eqs. (1) and (2). Once the eigenfunctions and eigenenergies are calculated, it is fairly straightforward to calculate the Lyapunov exponent¹⁷ $\gamma(E)$ and density of states $D(E)$ by directly using their definitions:

$$D(E) = \sum_j \delta(E - E_j) \quad (37)$$

and

$$\gamma(E_i) = \frac{1}{N-1} \sum_{j \neq i} \ln |E_j - E_i| = \frac{1}{N} \sum_{n=1}^N \ln \left| \frac{U_{n+1}}{U_n} \right|, \quad (38)$$

where E_j is the eigenenergy spectrum and N is the number of sites in the system. The Lyapunov exponent is the inverse localization length $\gamma = \xi^{-1}$. Then $\gamma=0$ for an extended state and $\gamma \neq 0$ for a localized state. The density of states and the Lyapunov exponent are related by the equation¹⁷

$$\gamma(E) = \int dE' D(E') \ln |E' - E|. \quad (39)$$

Of particular interest is the behavior of the Lyapunov exponent at the mobility edge

$$\gamma(E) \sim |E - E_c|^\beta \quad (40)$$

in the localized regions of E . Similarly, the behavior of the density of states at the mobility edge can be written as

$$D(E) \sim |E - E_c|^{-\delta}. \quad (41)$$

The critical exponents β and δ are clearly related by the equation

$$\beta = 1 - \delta \text{ with } \delta > 0. \quad (42)$$

Another important quantity describing the localization properties of the eigenfunction of eigenenergy E is the dimensionless resistance obtained by the Landauer formula.^{18,19} The arithmetic-mean resistivity $\rho(E)$ is defined as

$$\rho(E) = \frac{1}{N} \sum_{i=1}^N \frac{1}{N_i} R(E, N_i), \quad (43)$$

where N is the length of the system studied, and $R(E, N_i)$ is the resistance for an eigenenergy E and length N_i .

TABLE I. The location of the mobility edges of model (2) for different λ and $Q = \frac{1}{3}$.

λ	E_1	E_2	E_3
0.20	0.900 000 000 000	1.106 438 241 627	2.006 438 241 627
0.40	0.800 000 000 000	1.224 807 680 927	2.024 807 680 927
0.60	0.700 000 000 000	1.353 672 503 740	2.053 672 503 740
0.80	0.600 000 000 000	1.491 647 286 717	2.091 647 286 717
1.00	0.500 000 000 000	1.637 458 608 818	2.137 458 608 818
1.20	0.400 000 000 000	1.789 974 874 213	2.189 978 874 213
1.40	0.300 000 000 000	1.948 213 525 836	2.248 213 525 835
1.60	0.200 000 000 000	2.111 334 438 750	2.311 334 438 749
1.80	0.100 000 000 000	2.278 626 204 439	2.378 626 204 439
2.00	0.000 000 000 000	2.449 489 742 783	2.449 489 742 783

A. Case I: $Q = \frac{1}{2}$

In most of our calculations, the number N of sites in the system is chosen to be 10 000. Our numerical results for $Q = \frac{1}{2}$ are presented in Figs. 1–6. In Fig. 1, we show our calculated $D(E)$ [Figs. 1(a) and 1(b)] and $\gamma(E)$ [Fig. 1(c)] for model (2) using the parameters $\lambda = 1.0$, $\pi\alpha = 0.2$, and $\nu = 0.7$. The average density of states shown in Fig. 1(b) is obtained by taking an average over the irrelevant variable α (in the range $0.2 < \alpha < 0.8$). From Fig. 1, we can see that there are two subbands of energy. This is because for $Q = \frac{1}{2}$ and $\alpha = 0$ the model (2) is an exact periodic potential with period 2. The energy band consists of two subbands due to the Bloch theorem and the two subbands are symmetric about $E = 0$. In every subband, there are two mobility edges. The mobility edges are located at $E = \pm 1.0$ and ± 2.0 . Eigenstates with $|E| < E_{c_1} = \lambda = 1.0$ and $|E| > E_{c_2} = 2.0$ are found to be localized whereas the states with $\lambda < |E| < 2$ are extended,

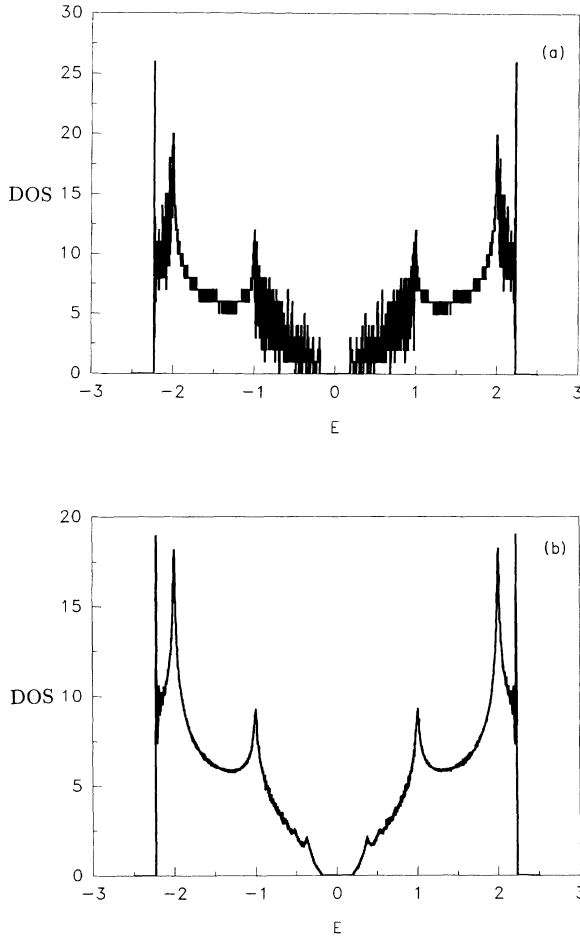


FIG. 1. (a) Density of states (DOS) for model (2) with parameters $Q = \frac{1}{2}$, $\nu = 0.70$, $\pi\alpha = 0.20$, and $\lambda = 1.0$. One can see that there are two subbands and a dramatic change when energy passes through the mobility edges. (b) Average DOS. The average is taken over the irrelevant variable α which is taken to be in the interval $[0.2, 0.8]$. (c) Lyapunov exponent γ vs energy. One can see the mobility edges are at $E_{c_1} = \pm 1.0$ and $E_{c_2} = \pm 2.0$.

in agreement with the theoretical results of Sec. II A.

In Fig. 2, we show the arithmetic-mean resistivity $\rho(E)$ as a function of N for four different eigenenergies using the same parameters as in Fig. 1. The curves 1, 2, 3, and 4 represent $\ln \rho$ vs N for eigenenergies $E_1 = 1.207\,075\,282\,591$, $E_2 = 0.963\,457\,961\,186$, $E_3 = 0.745\,571\,998\,299$, and $E_4 = 0.285\,951\,706\,605$, respectively. Corresponding to curve 1, the eigenenergy is E_1 , which is in the center of the subband. From Fig. 2, it can be seen that the resistivity exhibits features of an extended state (see curve 1). Corresponding to curves 2, 3, and 4, the eigenenergies are E_2 , E_3 , and E_4 , respectively. These eigenenergies are lower than the mobility edge $E_{c_1} = \lambda$, and the eigenstates are localized. For the localized state the value of the resistivity goes up as the site number N increases and finally approaches infinity. In particular, the curve 4 exhibits a more striking characteristic of the resistivity of the localized state. From Fig. 2, the value of the resistivity is seen to have reached 10^{280} even before the site position N reaches 1300 (see curve 4).

In Figs. 3(a)–3(d), we plot the corresponding eigen-

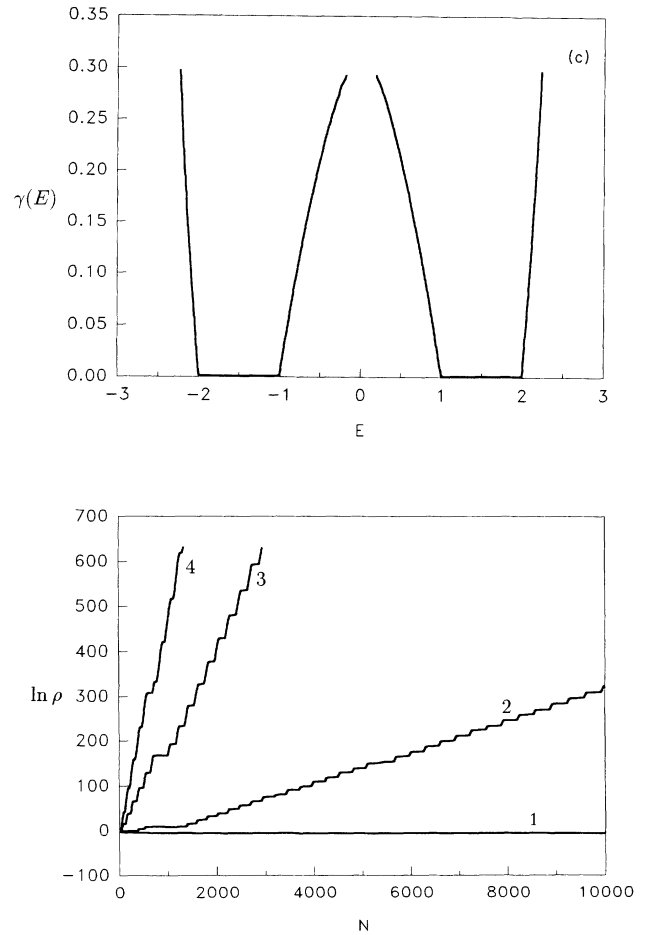


FIG. 2. Arithmetic-mean resistivities of the four different eigenenergies for model (2) using the same parameters as in Fig. 1, as a function of the site position N . Corresponding to curves 1, 2, 3, and 4, the eigenenergies are $E_1 = 1.207\,075\,282\,591$, $E_2 = 0.963\,457\,961\,186$, $E_3 = 0.745\,571\,998\,299$, and $E_4 = 0.285\,951\,706\,605$, respectively.

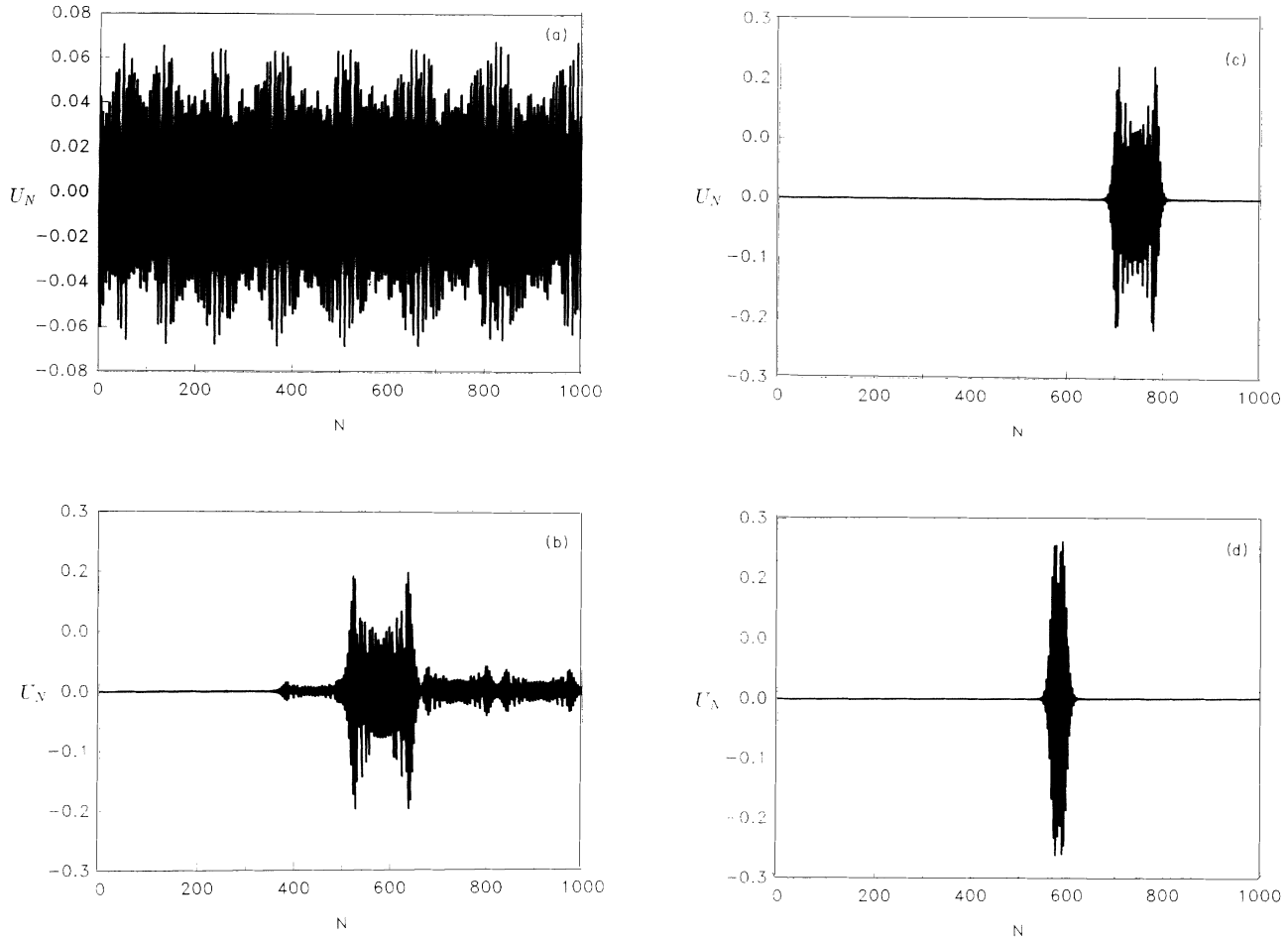


FIG. 3. Wave functions of the four eigenstates of Fig. 2. Corresponding to (a), (b), (c), and (d), the eigenenergies are $E_1=1.207\,075\,282\,591$, $E_2=0.963\,457\,961\,186$, $E_3=0.745\,571\,998\,299$, and $E_4=0.285\,951\,706\,605$, respectively. (a) A typical extended state; (b)–(d) localized states.

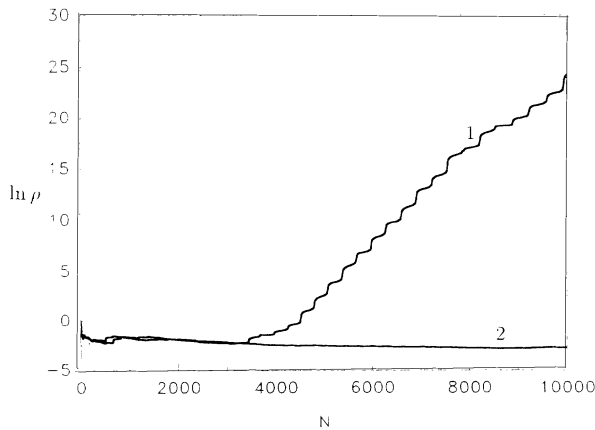


FIG. 4. Arithmetic-mean resistivities of the two eigenenergies which are near the mobility edge $E_{c_1}=1.0$ as a function of the site position N . Corresponding to curves 1 and 2, the energies are $E_1=0.997\,412\,398\,312 < E_{c_1}$ and $E_2=1.003\,830\,875\,513 > E_{c_1}$, respectively. One can see that indeed mobility edges exist.

states for the four eigenenergies of Fig. 2. Figure 3(a) shows the wave function of the extended state whose resistivity is given by curve 1 in Fig. 2. The wave function has the specific features of an extended state. In Figs. 3(b)–3(d), we show the wave functions of the localized states whose resistivity is given by the curves 2, 3, and 4 in Fig. 2, respectively. Obviously, the wave functions of these localized states are confined in a segment of the one-dimensional system and the localized states have more wiggles in them as one moves toward the mobility edges from the subband edge.

In Fig. 4, we show the arithmetic-mean resistivity for $E_1=0.997\,412\,398\,312 < E_{c_1}$ and $E_2=1.003\,830\,875\,513 > E_{c_1}$; E_1 and E_2 are very close the mobility edge $E_{c_1}=\lambda$. The asymptotic behavior of the resistivity for

TABLE II. The critical exponents for $Q=\frac{1}{2}$.

Mobility edge	$\pm\lambda$	± 2
β	0.9	1
δ	0.1	0

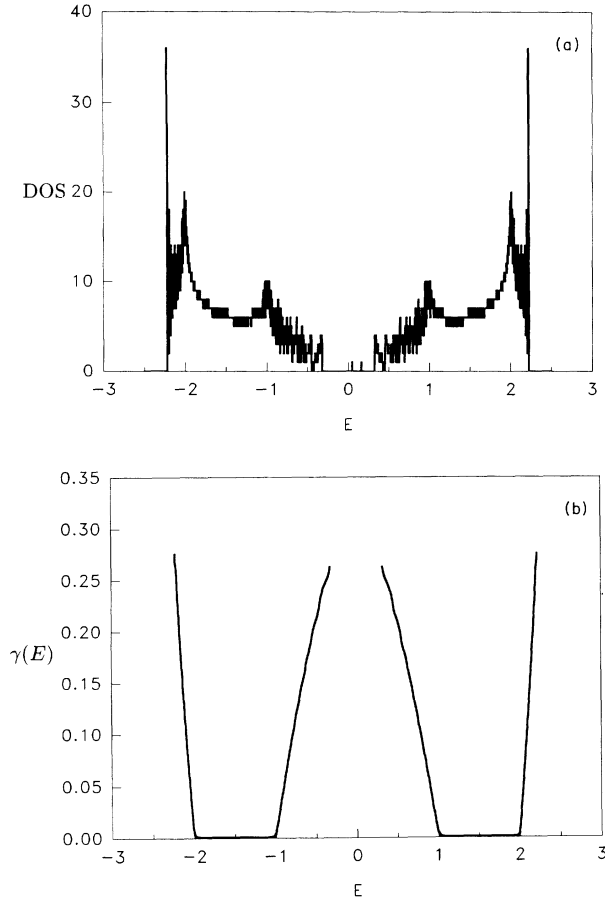


FIG. 5. (a) DOS for model (2) with parameters $Q = \frac{1}{2}$, $\nu = 0.70$, $\pi\alpha = 0.60$, and $\lambda = 1.0$. (b) Lyapunov exponent γ vs energy for the model defined in (a). The mobility edges are at $E_{c_1} = \pm 1.0$ and $E_{c_2} = \pm 2.0$.

very large systems shows the existence of the mobility edge at the $E = E_{c_1}$, with states just above E_{c_1} being extended so that $\rho(E)$ converges to zero in the limit $n \rightarrow \infty$, whereas the states just below E_{c_1} are localized with $\rho(E)$ approaching infinity as $n \rightarrow \infty$. Note that the closer one

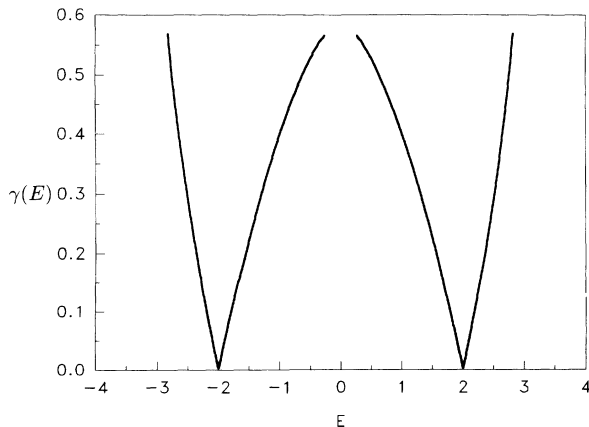


FIG. 6. Lyapunov exponent γ vs energy for model (2) with parameters $Q = \frac{1}{2}$, $\nu = 0.70$, $\pi\alpha = 0.20$, and $\lambda = 2.0$.

is in energy to the mobility edge, the longer the system must be for the infinite asymptotic behavior of the resistivity to show up.

We have explicitly calculated the critical exponents β and δ at mobility edges $E_{c_1} = \pm\lambda$ and $E_{c_2} = \pm 2$. The results are shown in Table II. At mobility edge $E_{c_2} = \pm 2$, the critical exponents are the same as those of the model

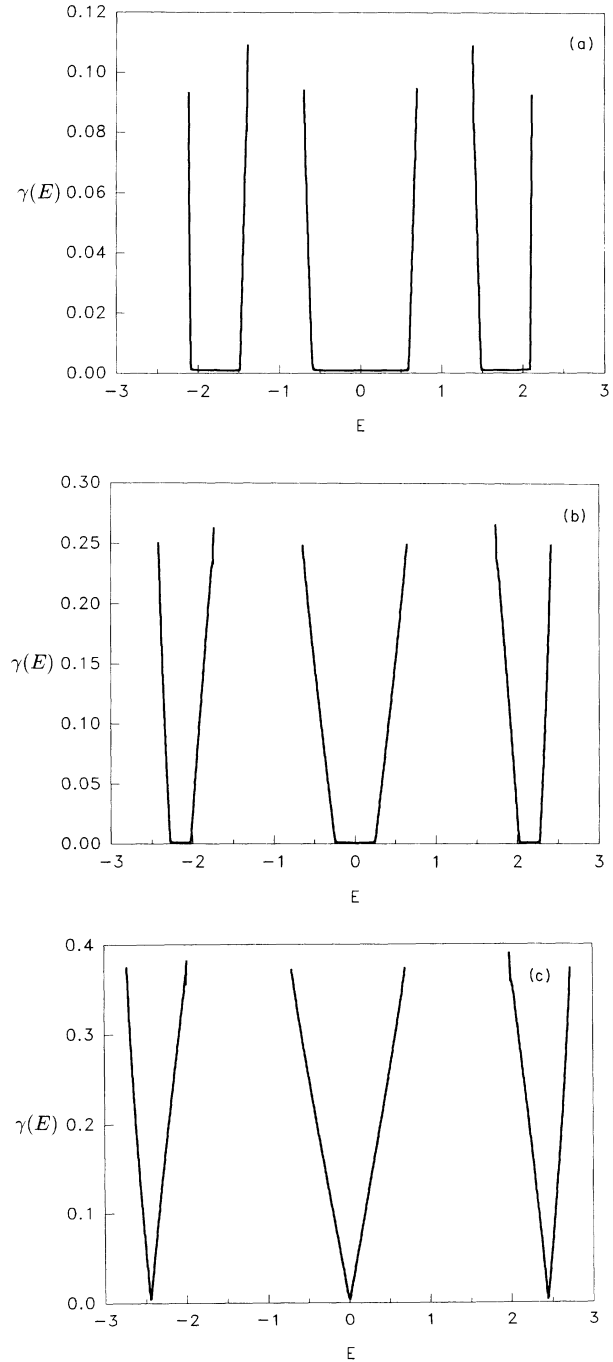


FIG. 7. Lyapunov exponent γ vs energy for model (2) with parameters $Q = \frac{1}{3}$, $\nu = 0.70$, $\pi\alpha = 0.20$, and (a) $\lambda = 0.8$, (b) $\lambda = 1.5$, and (c) $\lambda = 2.0$, respectively. One can see that there are three subbands and the mobility edges are the same as the values listed in Table I.

studied by Das Sarma, He, and Xia,¹⁵ but the critical exponents at mobility edge $E_{c_1} = \pm\lambda$ are different.

From the theoretical results described in Sec. II, we find that the mobility edges are only dependent on the parameters Q and λ . In order to check this conclusion, we have varied the parameters α and ν in our numerical work to ensure that the α, ν are indeed "irrelevant" variables. We find that the mobility edges and the critical exponents β and δ do not depend on the actual values of the parameters α and ν . In Fig. 5, we show our calculated density of states [Fig. 5(a)] and the Lyapunov exponent [Fig. 5(b)] for a value of $\pi\alpha=0.6$ different from that ($\pi\alpha=0.2$) used in Fig. 1. The other parameters have been kept the same. As one can see, the mobility edges, β , and δ do not depend on the irrelevant parameter α . The same is true of the irrelevant parameter ν . These results are same as the conclusion obtained by Das Sarma, He, and Xia¹⁵ for model (2) with $Q=0$.

In Fig. 6, we show our calculated $\gamma(E)$ by changing the value of the parameter λ to $\lambda=2$, keeping the other parameters the same ($\pi\alpha=0.2$, $\nu=0.7$) as in Fig. 1. In every subband, the mobility edges $E_{c_1} = \pm\lambda$ now merge and shift to other mobility edges $E_{c_2} = \pm 2$. When $\lambda=2$ the two mobility edges in every subband are equal and for $\lambda > 2$ all states are localized.

B. Case II: $Q = \frac{1}{3}$

In our calculation for $Q = \frac{1}{3}$, the method of calculation and the number of sites in the system are same as in Sec. III A. Our numerical results are shown in Figs. 7(a)–7(c). In the Figs. 7(a)–7(c), we show the Lyapunov exponent $\gamma(E)$ as a function of eigenenergy E for $\lambda=0.8, 1.5$, and 2.0 , respectively. The other parameters are $\pi\alpha=0.2$ and $\nu=0.7$. From Fig. 7, we can see that the energy band consists of three subbands for every λ . This is because, for $Q = \frac{1}{3}$ and $\alpha=0$, the model (2) is an exact periodic potential with period 3 and the energy band consists of three subbands due to the Bloch theorem. There are two mobility edges in every subband with localized states at the edge of the subband and extended states in the center of the subband. The mobility edges from the numerical results are in excellent agreement with the values listed in Table I. We have explicitly calculated the critical exponents β and δ at every mobility edge. We find that the critical exponents of $Q = \frac{1}{3}$ are more complex than those of $Q = \frac{1}{2}$ and 0. The critical exponents depend not only on the mobility edge but also on the parameter λ . Simi-

larly, we carry out the same calculation as in Sec. III A for $Q = \frac{1}{2}$ and get the same results, such as that α, ν are irrelevant parameters and all states are localized for $\lambda > 2$, etc.

IV. CONCLUSIONS

In this paper, we have studied the localization properties of a class of nearest-neighbor one-dimensional tight-binding models where the on-site diagonal potential is defined by Eq. (2). This potential is an asymptotic periodic function. By using some simple arguments, we find that there are localized and extended states and mobility edges in these systems, and get the values of the mobility edge. We explicitly verify our theoretical predictions by numerical simulation on the model. The interesting features of the model are the following.

(1) It is clear that all the numerical results are in excellent agreement with the theoretical results developed in Sec. II for describing the localization properties of the class of one-dimensional asymptotic periodic potentials being studied in this paper. We can conclude that the predictive ability of the theory is good for model (2).

(2) The conclusion from the Lyapunov exponent through the Thouless formula is in accordance with that of the resistivity. This implies that the Lyapunov exponent is a good index for describing the localization properties of the wave function for model (2).

(3) The features of the energy band only depend on the parameter Q . There are two and three subbands for $Q = \frac{1}{2}$ and $\frac{1}{3}$, respectively. Generally, for $Q = n/m$, we can conclude that the energy band consists of m subbands.

(4) When $\lambda < 2$, there are two mobility edges in every subband with the extended states in the center of the subband and the localized states at the edge of the subband, and all states are localized for $\lambda > 2$.

(5) For $Q \neq 0$, the critical exponents are more complex than those for $Q=0$ and may depend on the parameter λ and the mobility edges.

(6) The parameters α and ν are indeed irrelevant parameters.

Finally, we point out that we do not discuss the case of irrational Q in this paper. When Q is irrational, the potential defined by model (2) is an asymptotic periodic function with period incommensurate with the period of the lattice. This is an asymptotic quasiperiodic function. We will study this case in another paper.

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